

BestOf

Amazon reviews into clear buying advice

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The pitch

Messy reviews in. Clear advice out.

- BestOf turns thousands of Amazon reviews into a short buying guide.
- Upload a CSV, process the reviews, choose a category, and get the best picks.
- The user saves time and still sees the reason behind the recommendation.

57,418

clean reviews

125

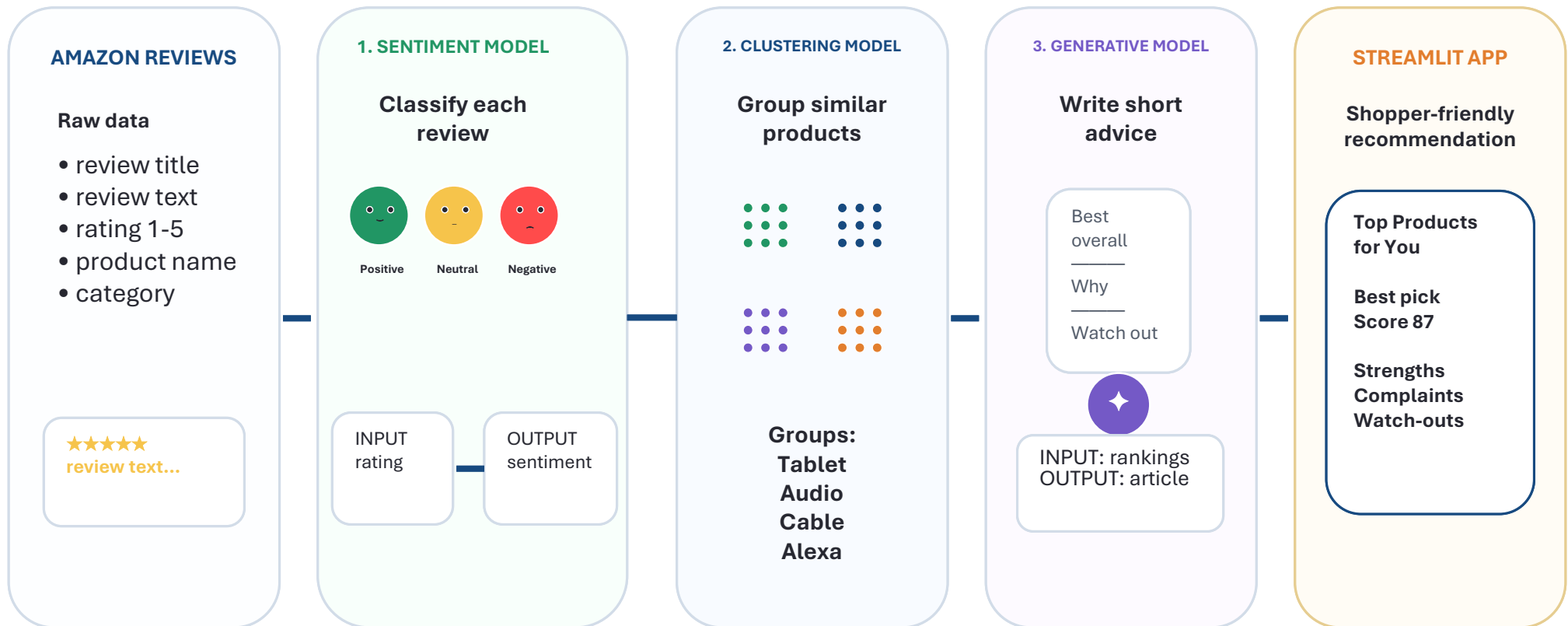
products

6

groups

BestOf pipeline

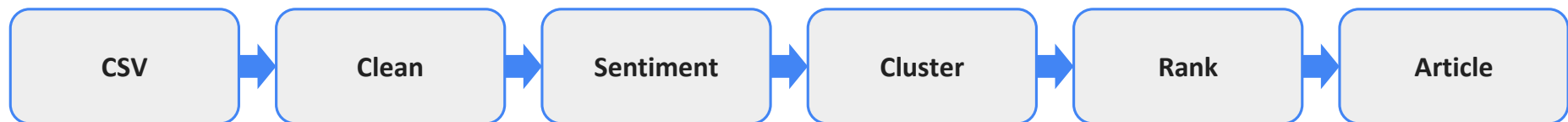
A simple flow from raw reviews to a shopper-friendly recommendation.



Methods

Simple methods, connected end to end.

- Dataset: Datafiniti Amazon reviews, 57,418 usable rows.
- Main columns: product name, review text, rating, categories.
- Ranking combines rating, positive share, and review volume.



Ranking score = rating + sentiment + review volume

Model 1 - Sentiment

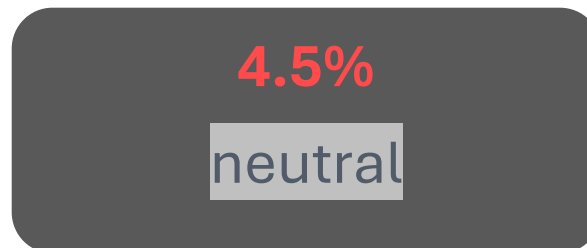
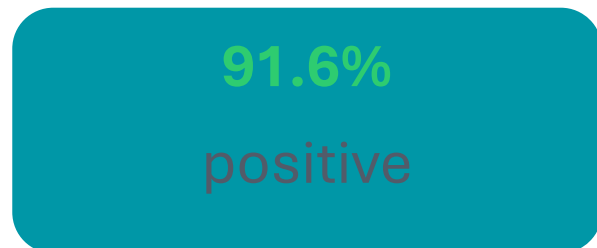
Sentiment starts with a clear rule.

- 1-2 stars = negative
- 3 stars = neutral
- 4-5 stars = positive
- Why this works here: it is transparent, fast, and enough for product ranking.

Sentiment result

The signal is useful, but not perfect.

- Most reviews are positive, which matches consumer-product review data.
- Neutral and negative reviews are still important because they reveal common complaints.
- Limitation: ratings can miss sarcasm or mixed opinions.



Model 2 - Clustering

Clustering makes the catalogue understandable.

- TF-IDF turns product/category text into features.
- KMeans groups products into readable categories.
- The app uses 6 groups so the demo stays easy to explain.

Discovered groups

**Cable, Tablet, Speaker
Audio, Laptop Tablet, Bag
Audio Pockets, Remote
Voice Alexa**

Clustering result

The choice is practical, not overcomplicated.

- Silhouette scores were checked across different k values.
- $k = 6$ gives a good balance between quality and readability.
- The groups are understandable enough for the buying-guide flow.

$k = 6$

chosen for demo

Model 3 - Article

The final output is human-readable advice.

- FLAN-T5 writes a short article from category insights.
- It receives aggregated facts only: top products, scores, and complaints.
- A deterministic fallback keeps the app reliable if generation is weak.

Example article output

**Best overall: Amazon 9W
PowerFast charger**

**Why: strong rating, high positive
share, enough reviews**

**Watch out: charge, app, battery,
screen, and wifi issues**

Article quality

The goal is useful output, not fancy text.

- Each category gets one article.
- The article must include a best pick, alternatives, warnings, and a bottom line.
- The fallback makes the demo stable even when the LLM is slow.

6

articles

0

raw reviews sent

Yes

fallback

Takeaway

A simple product built from several small parts.

- BestOf is an end-to-end ML product.
- It connects data cleaning, sentiment, clustering, ranking, text generation, API, UI, Docker, and EC2.
- The main value: the user gets a recommendation they can understand.

Streamlit

FastAPI

Docker

EC2

Hugging Face

Demo

Three minutes: from upload to recommendation.

1. Open the deployed Streamlit app.
2. Process the review CSV.
3. Pick a category.
4. Generate the buying guide.

Live deployment on AWS

<http://18.157.233.122:8501/>